

Characteristics of Wireless Transmission

- **Wireless networks (WLANs)**
 - **WLANs** send signals through the **air** using radio frequency (RF) waves.
- Wired and wireless signals are similar because they both use Layer 3 and higher protocols.
 - Wireless transmission is affected by the atmosphere, making it different from wired transmission.

The Wireless Spectrum

- **The wireless spectrum is a range of electromagnetic waves used for sending data and voice signals.**
 - It is ordered by frequency, from lowest to highest, and covers 9 KHz to 300 GHz.
- Wireless services are usually limited to a certain area.
- The **FCC** controls frequencies in the United States.
- The **ITU** manages international frequencies because signals can cross borders.



Antennas

- **Radiation pattern:** Shows signal strength in 3D space.
 - It includes all energy an antenna sends or receives.
- **Unidirectional antenna:** Sends wireless signals in one direction.
- **Omnidirectional antenna:** Sends and receives wireless signals in all directions with **equal** strength and clarity in all directions.
- **Range:** Area where the antenna signal can be received.



Signal Propagation(yayılmaması)

- **Line-of-sight (LOS)** means the signal moves in a straight line from the transmitter to the receiver.
- If something blocks the signal, it can:
 - **Pass through** the object.
 - **Be absorbed** by the object.
 - **Be affected** by three effects:
 - **Reflection:** the signal bounces back.
 - **Diffraction:** the signal splits into smaller waves.
 - **Scattering:** the signal spreads in different directions.



Multipath Signals – Summary

- Wireless signals can travel using **different paths** to reach the destination.
- This happens because of **reflection(yansımam), diffraction(kırınım), and scattering(saçılma)**.
- **Advantage:** Better chance of reaching destination
- **Disadvantage:** Signal delay will cause to data errors.

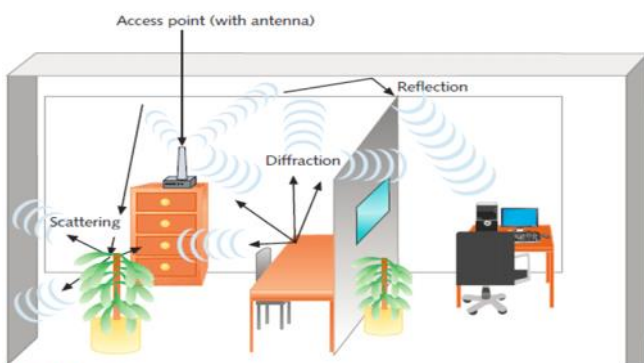


Figure 6-3 Multipath signal propagation

Frequency Ranges

- **2.4 GHz band** is older, has 11 unlicensed channels, but more interference.
- **5 GHz band** is newer, includes 24 unlicensed bands like 5.1–5.8 GHz, used also by radar and weather systems.
- **Unlicensed** means no FCC approval is needed.

Wireless PAN

- WPANs work in small areas and connect just a few personal devices.
- Common wireless options: **Bluetooth**, **Infrared (IR)**, and **NFC (Near-Field Communication)**.

Bluetooth

- Bluetooth works in the 2.4–2.485 GHz radio band.
- It switches between frequencies (**called frequency hopping**).
- Devices must be close and paired before connecting.
- **Bluejacking** sends messages without permission.
- **Bluesnarfing** downloads data without permission.

NFC (Near-Field Communication)

- NFC is a wireless technology for short-range communication.
- It sends data one way using tags, like for secure access.
- NFC tags get power from nearby devices through magnetic induction.

Wi-Fi WLAN(Wireless LAN) Architecture

no intervening connectivity device

- **Ad hoc WLAN**: Devices talk to each other **directly** using **wireless NICs** without a router or access point.
 - File sharing between two laptops
 - Mobile-to-mobile gaming connection
 - communication between war drones
- **Infrastructure WLAN**: Devices talk through **a wireless access point (WAP)**, not directly with each other. The access point needs **good power** and a **good location**.



Figure 6-10 An infrastructure WLAN

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Wi-Fi WLAN(Wireless LAN) Architecture

Mesh WLAN

- **Mesh WLAN** uses **multiple access points (APs)**
- It provide **stronger and more reliable** wireless connections.
- It provides **network fault tolerance**, meaning if one AP fails, others can keep the network working.



Figure 6-11 A mesh WLAN

- **Mobile networking** lets wireless devices move freely
 - Range depends on access method, equipment, and office layout
- **Wireless tech** can connect **two LANs**
 - Uses fixed link with directional antennas.
- Supports the **same protocols and operating systems** as wired LANs.

802.11 WLAN Standards

- **802.11** is a set of popular wireless standards created by **IEEE**.
- Common types include **802.11b, a, g, n, and ac**.
- All use **half-duplex**, meaning data is sent one direction at a time.
- These standards differ at the **Physical layer**.
- From **802.11n onwards**, they also change how frames are handled at the **MAC sublayer**.

Table 6-3 Technical details for 802.11 wireless standards

Standard	Frequency band	Max. theoretical throughput	Effective throughput	Geographic range
802.11b	2.4 GHz	11 Mbps	5 Mbps	100 m
802.11a	5 GHz	54 Mbps	11–18 Mbps	20 m
802.11g	2.4 GHz	54 Mbps	20–25 Mbps	100 m
802.11n	2.4 GHz or 5 GHz	65–600 Mbps	65–500 Mbps	400 m (if MIMO is used)
802.11ac	Wave 1 (3 data streams)	5 GHz	1.3 Gbps	Documented as high as 561 Mbps per client
	Wave 2 (4 data streams)	5 GHz	3.47 Gbps	(Unknown at time of printing)
	Wave 3 (8 data streams)	5 GHz	6.93 Gbps	(Unknown at time of printing)

- **802.11 MAC** adds a 48-bit physical address to frames.
 - This helps identify sender and receiver.
- It uses the **same addressing system** as other IEEE networks
 - Makes integration easier
- **Wireless devices:**
 - Can't send and receive at the same time
 - Can't fully avoid collisions
 - Use a **different access method** than Ethernet

ACCESS METHOD

- **CSMA/CA** is used to avoid data collisions
 - Uses **ACK** packets to confirm transmissions (ACK packets=Onay paketleri) verinin alınıp alınmadığını anlamak için
 - Adds more overhead than Ethernet (802.3) because it sends more control packages.
 - Real speed is less than the theoretical limit
- **RTS/CTS** protocol helps large data packets RTS(Request to Send) CTS(Clear to Send)
 - first sending device send RTS signal ,then if the medium is clear device replies with CTS.
 - Prevents other devices from interrupting-collusion.
 - Useful for large transmission packets and managing network traffic.
 - But lowers overall 802.11 performance

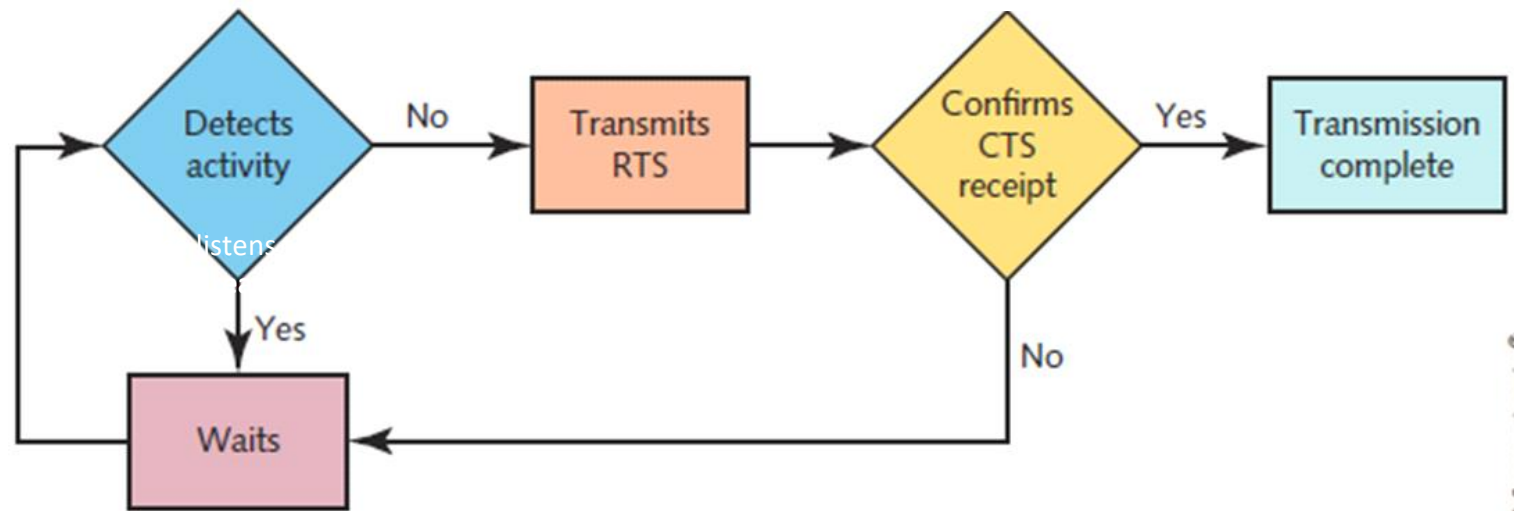


Figure 6-14 CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)

Association in WLAN: !

- A computer exchanges packets with an access point to connect to the internet.
- This is managed by the MAC sublayer.

Scanning:

- The device searches for available access points.
- **Active scanning**: sends a probe frame to find access points. (Yani "burada Wi-Fi var mı?" diye kendi sorar.)
- **Passive scanning**: waits and listens for beacon frames sent by access points. (Yani sessizce dinleyerek ağları tespit eder.)

- **SSID** = Ağın adı "Cafe_WiFi"
- **BSS** = Aynı ağa bağlı cihazların oluşturduğu grup
- **BSSID** = Bu grubun kimliği (genelde access point'in MAC adresi)

Association – SSID and BSS

- **SSID (Service Set Identifier)**
 - A unique name that identifies a wireless network.
 - Found in the **beacon** frame.
 - Set up on the access point.
 - Helps improve security and makes network management easier.

- **BSS (Basic Service Set)**
 - A group of devices using the same access point.
 - **BSSID** is the **unique ID of this group** (usually the MAC address of the access point).

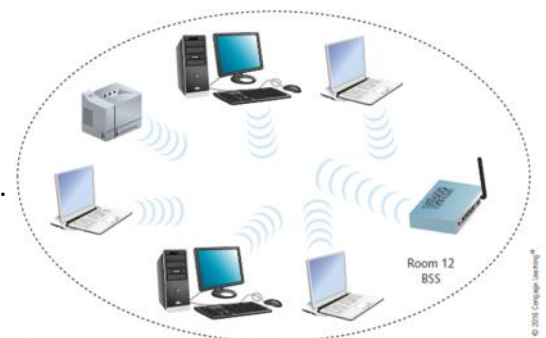


Figure 6-15 A network with a single BSS

ESS (Extended Service Set) is a group of multiple **BSSs** (Basic Service Sets) that are connected to the **same LAN** and work together as one larger wireless network.



- **ESS (Extended Service Set)** is a group of access points linked to the same LAN and using the same **ESSID()**.
 - It allows **roaming(dolaşım)**, meaning a device can move between BSSs without losing connection.
 - When multiple access points are available, the device connects to the **strongest and most reliable signal**.
 - However, this can be a **security risk** because the device might connect to a **rogue(fake) access point** pretending to be trusted.
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- "first-floor ESS" contains 3 different BSS(Room12-13-14).Every room has Access point(AP).
 - For example, if you're connected to Wi-Fi in **Room 12** and walk to **Room 14**, your device automatically switches to the access point in Room 14 **without losing connection**. This is called **roaming**.

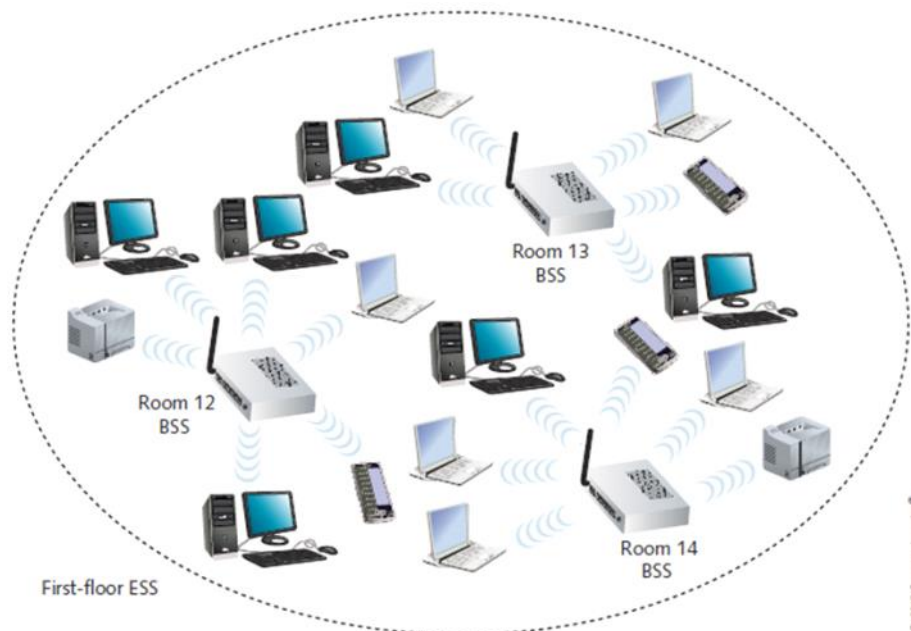


Figure 6-16 A network with multiple BSSs forming an ESS

- In an **ESS**, a device can connect to any access point **while staying online**.
- **Reassociation** happens when a user moves from one access point's range to another.
 - This can cause errors during the switch.
- **Scanning** helps devices balance the network load by choosing the best access point.

Roaming = User-level behavior

Reassociation = The technical process done by the device

802.11 Wireless Network Security

- **Default Security:**
 - No security by default; only SSID is needed to connect.
- **Authentication:**
 - Verifies client credentials by checking them in a database.
- **MAC Filtering:**
 - Blocks devices not listed by their MAC address from connecting.
- **Encryption:**
 - Uses algorithms to scramble data for secure transmission.

WPA/WPA2

- **WPA:**
 - Gives a different key to each transmission for better security.
- **WPA2:**
 - Improved version of WPA with stronger encryption.
- **WPA/WPA2-Enterprise:**
 - Combines with a RADIUS server for the highest level of security.

Summary of Wireless Concepts

- Most wireless networks operate at **2.4 GHz or 5 GHz** frequencies.
- Signals can **pass through, be absorbed, or be affected** by reflection, diffraction, or scattering.
- **PANs** (Personal Area Networks) are short-range and usually connect only a few personal devices.
- **Bluetooth** helps connect mobile devices and computers using a **shared communication standard**.